# Module: 2, Random Variable (Mathematical Expectation)

# Mean of a Random Variable (Mathematical Expectation)

Assuming that two fair coins was tossed, we find that the sample space for our experiment is $S = \{HH, HT, TH, TT\}$. Denote by $X$ the number of heads. Since the 4 sample points are all equally likely, it follows that $P(X = 0) = P(TT) = \frac{1}{4}, P(X = 1) = P(TH) + P(HT) = \frac{1}{2}$ , and $P(X = 2) = P(HH) = \frac{1}{4}$ , where a typical element, say $TH$, indicates that the first toss resulted in a tail followed by a head on the second toss. Now, these probabilities are just the relative frequencies for the given events in the long run. Therefore, Mean$= (0)\frac{1}{4} + (1)\frac{1}{2} + (2)\frac{1}{4} = 1$.

This result means that a person who tosses 2 coins over and over again will, on the average, get 1 head per toss.

## Definition 1

Let $X$ be a random variable with probability distribution. The mean, or expected value, of $X$ is

$$E(X) = \begin{cases} \sum_{x \in X} xP(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} xf(x)dx & \text{if } X \text{ is continuous.} \end{cases}$$

In general $E(X^r) = \begin{cases} \sum_{x \in X} x^r P(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x^r f(x)dx & \text{if } X \text{ is continuous.} \end{cases}$

• $Var(X) = E(X^2) - [E(X)]^2$

## Problem 1

*Find the expected value of $X$.*

| X | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| P(x)=P(X=x) | 1/35 | 12/35 | 18/35 | 4/35 |

**Solution:**

$$E(x) = 0(1/35) + 1(12/35) + 2(18/35) + 3(4/35) = 12/7.$$

## Problem 2

*Let $X$ be the random variable that denotes the life in hours of a certain electronic device. The probability density function is*

$$f(x) = \begin{cases} \frac{20000}{x^3} & \textit{if } x > 100 \\ 0 & \textit{otherwise} \end{cases}$$

*Find the expected life of this type of device.*

**Solution:**

$$E(X) = \int_{-\infty}^{\infty} (x) f(x) dx = \int_{100}^{\infty} \frac{20000}{x^2} dx = 200.$$

Therefore, we can expect this type of device to last, on average, 200 hours.

Theorem: Let $X$ be a random variable with probability distribution. The expected value of the random variable $Y = g(X)$ is

- $E(Y = g(X)) = \sum g(X)P(X = x)$, if $X$ is discrete, and
- $E(Y = g(X)) = \int_{-\infty}^{\infty} g(X)f(x)dx$, if $X$ is continuous.

## Problem 3

Suppose that the number of cars X that pass through a car wash between 4:00 P.M. and 5:00 P.M. on any sunny Friday has the following probability distribution:

| X | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|
| P(X=x) | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{1}{6}$ | $\frac{1}{6}$ |

Let $Y = 2X - 1$ represent the amount of money, in INR (Rs), paid to the attendant by the manager. Find the attendant's expected earnings for this particular time period.

# Solution

By the above Theorem, the attendant can expect to receive

$$
\begin{aligned}
E(Y = 2X - 1) &= \sum_{x=4}^{x=9} (2x - 1)P(X = x) \\
&= (7)\left(\frac{1}{12}\right) + (9)\left(\frac{1}{12}\right) + (11)\left(\frac{1}{4}\right) + (13)\left(\frac{1}{4}\right) \\
&+ (15)\left(\frac{1}{6}\right) + (17)\left(\frac{1}{6}\right) \\
&= \text{Rs.}12.67
\end{aligned}
$$

## Problem 4

Let $X$ be a random variable with density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2 \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of $Y = 4X + 3$.

Solution:

By the above Theorem,

$$E(Y = 4X + 3) = \int_{-1}^{2} \frac{(4x+3)(x^2)}{3} dx = \int_{-1}^{2} \frac{1}{3}(4x^3 + 3x^2) dx = 8.$$

# Mathematical Expectation Properties

- $E[(aX+b)] = aE(x) + b$
- $Var[(aX+b)] = a^2 Var(X)$
- The variables $X$ and $Y$ are independent if $E(XY) = E(X)E(Y)$
- $Cov(X,Y) = E(XY) - E(X)E(Y)$
- Correlation between $X$ and $Y$ is $\rho(X,Y) = \frac{Cov(X,Y)}{\sigma_x \sigma_y}$, where $\sigma_x$ is a standard deviation of $X$ and $\sigma_y$ is a standard deviation of $Y$.
- If the variables $X$ and $Y$ are independent, then $Cov(X,Y) = E(XY) - E(X)E(Y) = 0$ and hence $\rho(X,Y) = 0$.
- $-1 \leq \rho(X,Y) \leq 1$.

•$\sigma_x = \sqrt{Var(X)}$ and $\sigma_y = \sqrt{Var(Y)}$

# Definition

Let $X$ and $Y$ be random variables with joint probability distribution. The mean, or expected value, of the random variable $Z = g(X, Y)$ is

- $E(Z) = \sum_{X=x} \sum_{Y=y} g(X, Y)p(x, y)$, if $(X, Y)$ is discrete, and
- $E(Z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(X, Y)f(x, y)dxdy$, if $(X, Y)$ is continuous.

# Note

If $g(X, Y) = X$ in the above Definition, we have

$$E(X) = \begin{cases} \sum_{X=x} \sum_{Y=y} xp(x, y) \quad = \sum_{X=x} xP_x(x), & \text{discrete case,} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xf(x, y)dydx = \int_{-\infty}^{\infty} xf_X(x)dx, & \text{continuous case.} \end{cases}$$

where $P_X(x)(f_X(x))$ is the marginal distribution of $X$(marginal density function of the random variable $X$) Therefore, in calculating $E(X)$ over a two-dimensional space, one may use either the joint probability distribution of $X$ and $Y$ or the marginal distribution of $X$. (marginal density function of the random variable $x$).

# Note

Similarly, we define If $g(X, Y) = Y$ in the above Definition, we have

$$E(Y) = \begin{cases} \sum\limits_{X=x} \sum\limits_{Y=y} yp(x, y) &= \sum\limits_{Y=y} yP_Y(y), \quad \text{discrete case,} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} yf(x, y)dxdy &= \int_{-\infty}^{\infty} yf_Y(y)dy, \quad \text{continuous case.} \end{cases}$$

where $P_Y(y)(f_Y(y))$ is the marginal distribution of the random variable $Y$ (marginal density function of the random variable $Y$).

The expectation of $X^2$ is

$$E(X^2) = \begin{cases} \sum\limits_{X=x}\sum\limits_{Y=y} x^2 p(x,y) & = \sum\limits_{X=x} x^2 P_x(x), \quad \text{discrete case,} \\ \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x^2 f(x,y)dydx & = \int_{-\infty}^{\infty} x^2 f_X(x)dx, \quad \text{continuous case.} \end{cases}$$

Similarly, we can define $E(Y^2)$. In generally, we can define $E(X^r), r \geq 1$ and $E(Y^r), r \geq 1$. for the discrete and continuous random variables $X$ and $Y$.

## Problem 5

Let $X$ and $Y$ be the random variables with joint probability distribution indicated in the following Table.

| | $p(x,y)$ | x | | |
|---|---|---|---|---|
| | | 0 | 1 | 2 |
| y | 0 | $\frac{3}{28}$ | $\frac{9}{28}$ | $\frac{3}{28}$ |
| | 1 | $\frac{3}{14}$ | $\frac{3}{14}$ | 0 |
| | 2 | $\frac{1}{28}$ | 0 | 0 |

Find $E(X), E(Y)$ and $E(XY)$.

# Solution

The marginal distribution function of $X$ and $Y$ is

| | $p(x,y)$ | x | | | $P_Y(y)$ |
|---|---|---|---|---|---|
| | | 0 | 1 | 2 | |
| $y$ | 0 | $\frac{3}{28}$ | $\frac{9}{28}$ | $\frac{3}{28}$ | $\frac{15}{28}$ |
| | 1 | $\frac{3}{14}$ | $\frac{3}{14}$ | 0 | $\frac{3}{7}$ |
| | 2 | $\frac{1}{28}$ | 0 | 0 | $\frac{1}{28}$ |
| $P_X(x)$ | | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ | 1 |

To find $E(X)$ and $E(Y)$ using joint probality mass function and marginal distribution of $X$ and $Y$ (Exercise.)

# Solution Cont...

$$E(XY) = \sum_{x=0}^{x=2}\sum_{y=0}^{y=2} xyp(x,y)$$

$$= (0)(0)p(0,0) + (0)(1)p(0,1) + (0)(2)p(0,2)$$

$$+ (1)(0)p(1,0) + (1)(1)p(1,1) + (1)(2)p(1,2)$$

$$+ (2)(0)p(2,0) + (2)(1)p(2,1) + (2)(2)p(2,2)$$

$$= 0 + 0 + 0 + 0 + p(1,1) + 2p(1,2) + 0 + 2p(2,1) + 4p(2,2)$$

$$= 0 + \frac{3}{14} + (2)(0) + 0 + (2)(0) + 4(0)$$

$$= \frac{3}{14}.$$

## Problem 6

Find $E\left(\frac{Y}{X}\right)$ for the density function

$$f(x,y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 < x < 2, 0 < y < 1 \\ 0, & \text{elsewhere.} \end{cases}$$

Solution:We have

$$\begin{aligned} E\left(\frac{Y}{X}\right) &= \int_0^2 \int_0^1 \left(\frac{y}{x}\right) f(x,y)dxdy \\ &= \int_0^1 \int_0^2 \left(\frac{y(1+3y^2)}{4}\right) dxdy \\ &= \int_0^1 \left(\frac{y+3y^3}{2}\right) dy = \frac{5}{8}. \end{aligned}$$

## Problem 7

The joint probability mass function of two dimensional random variables $(X, Y)$ is $p(x, y) = \frac{x+y}{21}, x = 1, 2, 3$ and $y = 1, 2.$

1. Find the mean of $X$ and $Y$
2. Find the variance of $X$ and $Y$
3. Find the standard deviation of $X$ and $Y$.
4. Find the covariance of $X$ and $Y$.
5. Find the correlation bewteen $X$ and $Y$.

# Solution

The joint probability mass function of two dimensional random variables $(X, Y)$ is

| | $p(x,y)$ | $y$ = 1 | $y$ = 2 |
|---|---|---|---|
| $x$ | 1 | $\frac{2}{21}$ | $\frac{3}{21}$ |
| | 2 | $\frac{3}{21}$ | $\frac{4}{21}$ |
| | 3 | $\frac{4}{21}$ | $\frac{5}{21}$ |

# Solution Cont...

$$\text{For}(1),\ E(X) = \sum_{x=1}^{3}\sum_{y=1}^{2} xp(x,y)$$

$$= (1)p(1,1) + (1)p(1,2) + (2)p(2,1) + (2)p(2,2)$$

$$+ (3)p(3,1) + (3)p(3,2)$$

$$= (1)\left(\frac{2}{21}\right) + (1)\left(\frac{3}{21}\right) + (2)\left(\frac{3}{21}\right) + (2)\left(\frac{4}{21}\right)$$

$$+ (3)\left(\frac{4}{21}\right) + (3)\left(\frac{5}{21}\right)$$

$$= \frac{46}{21}.$$

# Solution Cont...

$$
\begin{aligned}
E(Y) &= \sum_{x=1}^{3}\sum_{y=1}^{2} yp(x,y) \\
&= (1)p(1,1) + (1)p(2,1) + (1)p(3,1) \\
&+ (2)p(1,2) + (2)p(2,2) + (2)p(3,2) \\
&= (1)\left(\frac{2}{21}\right) + (1)\left(\frac{3}{21}\right) + (1)\left(\frac{4}{21}\right) \\
&+ (2)\left(\frac{3}{21}\right) + (2)\left(\frac{4}{21}\right) + (2)\left(\frac{5}{21}\right) \\
&= \frac{33}{21}.
\end{aligned}
$$

# Solution Cont...

$$\text{For}(2), E(X^2) = \sum_{x=1}^{3}\sum_{y=1}^{2} x^2 p(x,y)$$

$$= (1^2)p(1,1) + (1^2)p(1,2) + (2^2)p(2,1) + (2^2)p(2,2)$$

$$+ (3^3)p(3,1) + (3^2)p(3,2)$$

$$= (1)\left(\frac{2}{21}\right) + (1)\left(\frac{3}{21}\right) + (4)\left(\frac{3}{21}\right) + (4)\left(\frac{4}{21}\right)$$

$$+ (9)\left(\frac{4}{21}\right) + (9)\left(\frac{5}{21}\right)$$

$$= \frac{114}{21}.$$

## Solution Cont...

$$
\begin{aligned}
E(Y^2) &= \sum_{x=1}^{3}\sum_{y=1}^{2} y^2 p(x,y) \\
&= (1^2)p(1,1) + (1^2)p(2,1) + (1^2)p(3,1) \\
&+ (2^2)p(1,2) + (2^2)p(2,2) + (2^2)p(3,2) \\
&= (1)\left(\frac{2}{21}\right) + (1)\left(\frac{3}{21}\right) + (1)\left(\frac{4}{21}\right) \\
&+ (4)\left(\frac{3}{21}\right) + (4)\left(\frac{4}{21}\right) + (4)\left(\frac{5}{21}\right) \\
&= \frac{57}{21}.
\end{aligned}
$$

# Solution Cont...

$$
\begin{aligned}
\sigma_x^2 = \text{Var}(X) = E(X^2) - [E(X)]^2 \\
&= \frac{114}{21} - \left[\frac{46}{21}\right]^2 = \frac{2394}{441} - \frac{2116}{441} \\
&= \frac{278}{441}.
\end{aligned}
$$

$$
\begin{aligned}
\sigma_y^2 = \text{Var}(Y) = E(Y^2) - [E(Y)]^2 \\
&= \frac{57}{21} - \left[\frac{33}{21}\right]^2 = \frac{1197}{441} - \frac{1089}{441} \\
&= \frac{108}{441}.
\end{aligned}
$$

# Solution Cont...

$$\text{For}(3), \sigma_x = \sqrt{Var(X)} = \sqrt{\frac{278}{441}} = 0.794$$

$$\sigma_y = \sqrt{Var(Y)} = \sqrt{\frac{108}{441}} = 0.495$$

For(4), Cov(X,Y) $= E(XY) - E(X)E(Y)$.

First we have to find $E(XY)$.

# Solution Cont...

$$E(XY) = \sum_{x=1}^{3}\sum_{y=1}^{2} xyp(x,y)$$

$$= (1)(1)p(1,1) + (1)(2)p(1,2) + (2)(1)p(2,1) + (2)(2)p(2,2)$$

$$+ (3)(1)p(3,1) + (3)(2)p(3,2)$$

$$= (1)p(1,1) + (2)p(1,2) + (2)p(2,1) + (4)p(2,2)$$

$$+ (3)p(3,1) + (6)p(3,2)$$

$$= (1)\left(\frac{2}{21}\right) + (2)\left(\frac{3}{21}\right) + (2)\left(\frac{3}{21}\right) + (4)\left(\frac{4}{21}\right)$$

$$+ (3)\left(\frac{4}{21}\right) + (6)\left(\frac{5}{21}\right) = \frac{72}{21}.$$

# Solution Cont...

$$\text{Therefore,} \quad \text{Cov(X,Y)} = E(XY) - E(X)E(Y)$$
$$= \frac{72}{21} - \left(\frac{46}{21}\right)\left(\frac{33}{21}\right)$$
$$= \frac{1512}{441} - \frac{1518}{441}$$
$$= \frac{-6}{441}$$
$$= -0.0136$$

# Solution Cont...

$$\textcolor{red}{For(5),} \quad \rho(X,Y) = \frac{Cov(X,Y)}{(\sigma_x)(\sigma_y)}$$

$$= \frac{-0.0136}{(0.794)(0.495)}$$

$$= \frac{-0.0136}{0.393}$$

$$= -0.035$$

## Problem 8

Let $X$ and $Y$ be the random variables with joint probability distribution indicated in the following Table.

| | $p(x,y)$ | x | | |
|---|---|---|---|---|
| | | 0 | 1 | 2 |
| $y$ | 0 | $\frac{3}{28}$ | $\frac{9}{28}$ | $\frac{3}{28}$ |
| | 1 | $\frac{3}{14}$ | $\frac{3}{14}$ | 0 |
| | 2 | $\frac{1}{28}$ | 0 | 0 |

- Find the covariance of $X$ and $Y$.
- Find the correlation between $X$ and $Y$.

## Problem 9

The fraction $X$ of male runners and the fraction $Y$ of female runners who compete in marathon races are described by the joint density function

$$f(x,y) = \begin{cases} 8xy, & 0 \leq y \leq x \leq 1 \\ 0, & \text{elesewhere.} \end{cases}$$

1. Find the covariance of $X$ and $Y$.
2. Find the correlation between $X$ and $Y$.

# Solution

We first compute the marginal density functions of $X$ and $Y$(Exercise).

They are

$$f_X(x) = \begin{cases} 4x^3, & 0 \le x \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

$$f_Y(y) = \begin{cases} 4y(1-y^2), & 0 \le y \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

# Solution Cont...

From these marginal density functions, we compute

$E(X) = \int_0^1 x4x^3 dx = \frac{4}{5}$ and

$E(Y) = \int_0^1 y4y(1-y^2)dy = \frac{8}{15}$.

From the joint density function given above, we have

$E(XY) = \int_0^1 \int_y^1 xy8xydxdy = \frac{4}{9}$.

$$\textit{For}(1) \quad \text{Cov(X,Y)} = E(XY) - E(X)E(Y)$$

$$= \frac{4}{9} - \left(\frac{4}{5}\right)\left(\frac{8}{15}\right)$$

$$= \frac{4}{225}$$

# Solution Cont...

First we have to find $E(X^2), E(Y^2), \sigma_x^2$ and $\sigma_y^2$. $E(X^2) = \int_0^1 4x^5 dx = \frac{2}{3}$ and $E(Y^2) = \int_0^1 4y^3(1-y^2)dy = \frac{1}{3}$.

$$\begin{aligned}
\sigma_x^2 = \mathrm{Var(X)} &= E(X^2) - [E(X)]^2 \\
&= \frac{2}{3} - \left[\frac{4}{5}\right]^2 = \frac{2}{3} - \frac{16}{25} = \frac{2}{75}. \\
\sigma_y^2 = \mathrm{Var(Y)} &= E(Y^2) - [E(Y)]^2 \\
&= \frac{1}{3} - \left[\frac{8}{15}\right]^2 = \frac{1}{3} - \frac{64}{225} = \frac{11}{225}.
\end{aligned}$$

# Solution Cont...

$$\text{Therefore,} \quad \rho(X,Y) = \frac{\text{Cov(X,Y)}}{\sqrt{(\sigma_x^2)(\sigma_y^2)}}$$

$$= \frac{\frac{4}{225}}{\sqrt{\left(\frac{2}{75}\right)\left(\frac{11}{225}\right)}}$$

$$= \frac{4}{\sqrt{66}}.$$

# Moments

Let $X$ be a discrete random variable and $X$ takes the values

$x_1, x_2, x_3, \ldots, x_n$ with probabilities $p_1, p_2, p_3, \ldots, p_n$ Then $r^{th}$ moment is

- $\mu_r'$(about origin) $= \sum_{i=1}^{n} x_i^r p_i$
- $\mu_r'$(about any point x=A) $= \sum_{i=1}^{n} (x_i - A)^r p_i$
- $\mu_r'$(about mean $\bar{x}$) $= \sum_{i=1}^{n} (x_i - \bar{x})^r p_i$

Similarly, for continuous case on the interval $(a, b)$, the $r^{th}$ moment is

- $\mu_r'$(about origin) $= \int_a^b x^r f(x) dx$
- $\mu_r'$(about any point x=A) $= \int_a^b (x - A)^r f(x) dx$
- $\mu_r'$(about mean $\bar{x}$) $= \int_a^b (x - \bar{x})^r f(x) dx$

Mean$=E(X) = \mu_1'$ and
Variance $=\sigma^2 = E(X^2) - (E(X))^2 = \mu_2' - (\mu_1')^2$

# Moment Generating Function (MGF)

The expected values $E(X), E(X^2), E(X^3), \ldots,$ and $E(X^r)$ are called moments. As you have already experienced in some cases, the mean $\mu = E(X)$ and the variance $\sigma^2 = \mathrm{Var}(X)$, which are functions of moments, are sometimes difficult to find. Special functions, called moment-generating functions can sometimes make finding the mean and variance of a random variable simpler.

Although higher order moments of a random variable $X$ may be obtained directly by using the definition of $E(X^n)$, it will be easier in many problems to compute them through the characteristic function or equivalently through the moment generating function of the random variable $X$.

While the characteristic function analysis exists, the moment generating function need not . Moment Generating Function (MGF) of a random variable(discrete or continuous) $X$ is defined as $E(e^{tx})$, where $t$ is real variable, and denoted as $M_X(t)$.

If $X$ is discrete, then $M_X(t) = E(e^{tx}) = \sum_{i=1}^{n} e^{tx_i} p(x_i)$,
where $X$ takes the values $x_1, x_2, x_3, \ldots, x_n$ with probabilities $p_(x_1), p_(x_2), p_(x_3), \ldots p(x_n)$

If $X$ is continuous, then $M_X(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$.

# Properties of MGF

- The $r^{th}$ moment $\mu_r^{'}$ is the coefficient of $\frac{t^r}{r!}$ in the expansion of $M_X(t)$ in series of powers of $t$.

$$
\begin{aligned}
M_X(t) = E(e^{tx}) &= E\left[1 + \frac{tx}{1!} + \frac{(tx)^2}{2!} + \cdots + \frac{(tx)^r}{r!} + \dots\right] \\
&= 1 + tE(x) + \frac{t^2}{2!}E(x^2) + \cdots + \frac{t^r}{r!}E(x^r) + \dots \\
&= 1 + t\mu_1^{'} + \frac{t^2}{2!}\mu_2^{'} + \cdots + \frac{t^r}{r!}\mu_r^{'} + \dots
\end{aligned}
$$

Therefore, $\mu_r^{'}$ is the coefficient of $\frac{t^r}{r!}$ in the expansion of $M_X(t)$ in series of powers of $t$.

- $$\mu_r' = E(X^r) = \left[\frac{d^r}{dt^r}(M_X(t))\right]_{t=0}.$$

  In particular, if $r = 1$ then $\mu_1' = \left[\frac{d(M_X(t))}{dt}\right]_{t=0}$ is the mean of $X$.

- $M_{cX}(t) = M_X(ct)$
- $M_{aX+b}(t) = e^{bt}M_X(at)$
- $M_{X+Y}(t) = M_X(t)M_Y(t)$

# Characteristic function

Let $X$ be a discrete random variable and $X$ takes the values $x_1, x_2, x_3, \ldots$ with probabilities $p_1, p_2, p_3, \ldots$ Then the characteristic function is defined as $\phi_X(t) = E(e^{itx}) = \sum_r e^{itx_r} p(x_r)$.

Similarly, for continuous case $\phi_X(t) = E(e^{itx}) = \int_{-\infty}^{\infty} e^{itx} f(x) dx$.

# Properties of Characteristic function

- $\mu_r^{'} = E(X^r)$=the coefficient of $\frac{i^r t^r}{r!}$ in the expansion of $\phi_X(t)$ in series of ascending powers of $it$.
- $$\mu_r^{'} = E(X^r) = \frac{1}{i^r}\left[\frac{d^r}{dt^r}(\phi_X(t))\right]_{t=0}.$$

  In particular, if $r = 1$ then $\mu_1^{'} = \frac{1}{i}\left[\frac{d(\phi_X(t))}{dt}\right]_{t=0}$ is the mean of $X$.
- $\phi_{aX+b}(t) = e^{ibt}\phi(at)$
- $\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t)$

## Problem 10

Find the moment generating function of the random variable $X$ whose probability mass function $P(X = x) = \frac{1}{2^x}, x = 1, 2, 3, \ldots,$ Deduce the mean and variance from moment generating function.

Answer: $M_X(t) = \frac{e^t}{2-e^t}$ and mean is 2.

## Problem 11

*The density function of a continuous random variable $X$ is given by $f(x) = kx(2-x),\ 0 \le x \le 2$. Find $k$, MGF, mean and variance.*

## Problem 12

Find the characteristic function of the random variable $X$ whose probability mass function $P(X = r) = q^r p, r = 0, 1, 2, 3, \ldots$ and $p + q = 1$. Deduce the mean and variance from characteristic function.

Answer: $\phi_X(t) = p(1 - qe^{it})^{-1}$, mean is $\frac{q}{p}$ and variance is $\frac{q}{p^2}$.